

$\therefore$ The columns of F do not span R^5 . By the Invertible Matrix Theorem and the equivalence of the statements (h) and (d), ~~F~~ also contains a non-trivial solution.

$\therefore$ By Defn, $\text{Nul } F \neq \{0\}$ $\square$

32) If R is a 6×6 matrix and $\text{Nul } R$ is not the zero subspace, what can you say about $\text{Col } R$?

Solution: Given, $\text{Nul } R \neq \{0\}$. By Defn, $\text{Nul } R$ represents the set of all solns. to the eqn. $Rx=0$. $\therefore R \cdot 0=0$, we can conclude that the eqn. $Rx=0$ has a nontrivial solution. By the Invertible Matrix Theorem and the equivalence of the statements (d) and (h), we know that the columns of R do not span R^6 . $\therefore$ The set of all linear combinations of the columns of $R \neq R^6$ (Defn. of Span)

By Defn, $\text{Col } R \neq R^6$ $\square$

33) If A is a 4×4 matrix and $\text{Col } A = R^4$, what can you say about solutions of equations of the form $Ax=b$ for b in R^4 ?

Solution: By Defn of column space, $\text{Col } A$ is the set of all linear combinations of the columns of A . $\therefore$ By Defn, $\text{Col } A = \text{Span}\{\text{columns of } A\}$. $\therefore$ The columns of A span R^4 . By the Invertible Matrix Theorem and the equivalence of the statements (h) and (g), the equation $Ax=b$ has at least one solution for each b in R^4 .

$\therefore$ The statement a is equivalent to h and (g), A is invertible. By Theorem 5, $x=A^{-1}b$ is the unique solution, $Ax=b$ for each b in R^4 . $\square$

34) If P is a 5×5 matrix and $\text{Nul } P$ is the zero subspace, what can you say about solutions of equations of the form $Px=b$ for b in R^5 ?

Solution: Given, $\text{Nul } P = \{0\}$. By Defn. of nullspace, the matrix equation $Px=0$ has only the trivial solution. By the equivalence of the statements (a) and (d) from the Invertible Matrix Theorem, we know that P is an invertible matrix. By Theorem 5, for each b in R^5 , the matrix equation $Px=b$ has the unique solution $P^{-1}b$. $\square$